Solutions – Practice Problem Set 2

ECON 308 – International Economic Relations

Fall 2025

Short Conceptual

1. H–O: labour endowment increase. A higher L (holding K fixed) rotates the PPF outward toward the labour-intensive good. At given goods prices (small-country), by *Rybczynski*, output of the labour-intensive good rises and the capital-intensive good falls. With endogenous prices (large-country), the relative supply of the labour-intensive good rises, tending to reduce its relative price.

2. Rybczynski Theorem. Holding goods prices constant, an increase in one factor endowment increases output of the good using that factor intensively and reduces output of the other good (in a two-good, two-factor competitive economy with full employment).

3. Stolper–Samuelson in a labour-scarce country. If Country A is capital-scarce (so L relatively abundant), opening to trade raises the relative price of the labour-intensive good it exports. By Stolper–Samuelson, the real wage (return to the abundant factor) rises and the real rental rate (return to the scarce factor) falls.

4. Export- vs. import-biased growth. Export-biased growth expands the supply of the export good, tends to lower its world relative price, and *worsens* the home terms of trade (TOT). Import-biased growth reduces demand for imports (or raises supply of import-competing good), tends to raise the price of exports relative to imports, *improving* the TOT.

5. TOT improvement in the Standard Trade Model. An increase in $P_{\text{exports}}/P_{\text{imports}}$ pivots the trade line outward at the production point, allowing a higher indifference curve—so welfare rises in a distortion-free competitive economy.

6. External vs. internal economies. *External* economies: unit costs fall with *industry* output (many firms; imperfect competition not required), generating potential multiple equilibria and geographic concentration. *Internal* economies: unit costs fall with *firm* output, implying imperfect competition (e.g., monopolistic competition).

7. Path dependence. With external economies, an early, possibly accidental, concentration of production lowers unit costs, attracting more firms, further lowering costs. History ("accidents") can lock in an industrial location even if fundamentals are symmetric.

Analytical and Numerical Problems

Problem 1: Long-Run Factor Adjustment (H–O)

Data (Home): $K_H = 400,\ L_H = 800$. Sectoral factor intensities: textiles (labour-intensive) $K_T/L_T = 0.25$, steel (capital-intensive) $K_S/L_S = 2$.

(a) **Relative factor abundance.** Home's aggregate ratio is

$$\left(\frac{K}{L}\right)_H = \frac{400}{800} = 0.5.$$

Interpretation: By itself, 0.5 does not establish abundance; abundance is *relative across countries*. If Foreign has $(K/L)_F > 0.5$, then Home is **labour-abundant**; if $(K/L)_F < 0.5$, Home is **capital-abundant**. In what follows, we adopt the standard case that Home is labour-abundant (i.e., $(K/L)_F > 0.5$).

(b) **Export pattern.** By the H–O theorem, a labour-abundant country exports the labour-intensive good $\Rightarrow$ **Home exports textiles**.

(c) **Factor price effects (Stolper–Samuelson).** Opening to trade raises the relative price of the labour-intensive good that Home exports. Thus the *real wage* rises and the *real rental rate* falls (measured in terms of both goods).

(d) **10% rise in P_T (world price of textiles).** By Stolper–Samuelson magnification, w rises more than proportionately relative to the numéraire good used intensively in textiles, while r falls in real terms. Labour moves toward textiles in the new long-run equilibrium, and outputs reallocate accordingly.

Problem 2: Standard Trade Model and Welfare

PPF: $F^2 + 2E^2 = 200$; **world relative price:** $P_E/P_F = 2$.

(a) Value-maximizing production. Maximize $V = P_F F + P_E E$. Normalize $P_F = 1$, so $V = F + 2E$ subject to $F^2 + 2E^2 = 200$.
Lagrangian:

$$\mathcal{L} = F + 2E + \lambda\,(200 - F^2 - 2E^2).$$

First-order conditions (FOCs):

$$\frac{\partial \mathcal{L}}{\partial F} = 1 - 2\lambda F = 0 \ \Rightarrow\ F = \frac{1}{2\lambda}, \qquad \frac{\partial \mathcal{L}}{\partial E} = 2 - 4\lambda E = 0 \ \Rightarrow\ E = \frac{1}{2\lambda}.$$

Hence $F = E$ at the optimum. Substitute into the PPF:

$$F^2 + 2F^2 = 3F^2 = 200 \ \Rightarrow\ F = E = \sqrt{200/3} \approx 8.165.$$

$$\boxed{(F_P, E_P) = \left(\sqrt{\tfrac{200}{3}},\ \sqrt{\tfrac{200}{3}}\right) \approx (8.165,\ 8.165)}.$$

Production value: $V = F + 2E = 3F \approx 24.495$.

(b) Consumption with $F = E$ preferences and trade pattern. Consumption must satisfy the budget line $F + 2E = V$ at world prices and lie on the preference ray $E = F$. Thus $F + 2F = 3F = V \Rightarrow F_C = V/3 = \sqrt{200/3}$ and $E_C = F_C$. Therefore,

$$\boxed{(F_C, E_C) = (F_P, E_P)}.$$

Trade pattern: Zero trade volumes (the country happens to produce at its most-preferred bundle at these prices).

(c) Gains from trade vs. autarky. Because (F_P, E_P) already lies at the tangency of the world price line with the PPF *and* preferences pick $F = E$, autarky and free-trade consumption coincide. Hence there are no additional gains-from-trade at this price vector and preference ray (a knife-edge case). In general, with different preferences, trade would move consumption to a higher indifference curve.

(d) TOT improvement (rise in P_E/P_F). If electronics is the export good and P_E/P_F rises, the world price line becomes steeper, rotating outward at the production point. Firms re-optimize (more E, less F), and the budget line through the new production point supports a *higher* indifference curve. Thus welfare *increases*. (A standard STM diagram shows the new, steeper price line tangent to the PPF at a point with higher E/F, and a higher indifference curve.)

Problem 3: External Economies and Location

Costs and demand. Unit cost in a country with industry output Q:

$$C(Q) = 20 - 0.05Q, \quad \text{(external economies: costs fall with industry output).}$$

World demand: $Q^W = 400 - 5P \ \Rightarrow\ P = 80 - 0.2Q^W$.

(a) Price–quantity relationship. From demand:

$$\boxed{P = 80 - 0.2Q^W}.$$

(b) Symmetric production (each makes half). If A and B each produce $Q^W/2$, their unit costs are

$$C_A = C_B = 20 - 0.05\left(\frac{Q^W}{2}\right) = 20 - 0.025Q^W.$$

Under competition, P equals the common unit cost. Equate with inverse demand:

$$80 - 0.2Q^W = 20 - 0.025Q^W \ \Rightarrow\ 60 = 0.175\,Q^W \ \Rightarrow\ Q^W = \frac{60}{0.175} \approx 342.857.$$

Then

$$P = 80 - 0.2(342.857) \approx 11.4286, \quad C = 20 - 0.025(342.857) \approx 11.4286.$$

$$\boxed{Q^W \approx 342.86, \quad P \approx 11.43.}$$

(c) Positive feedback and specialization. Suppose A produces slightly more: $Q_A = \frac{Q^W}{2} + \varepsilon$, $Q_B = \frac{Q^W}{2} - \varepsilon$.

$$C_A = 20 - 0.05\left(\frac{Q^W}{2} + \varepsilon\right) = 20 - 0.025Q^W - 0.05\varepsilon,$$

$$C_B = 20 - 0.05\left(\frac{Q^W}{2} - \varepsilon\right) = 20 - 0.025Q^W + 0.05\varepsilon.$$

Hence

$$C_A - C_B = -0.1\,\varepsilon < 0 \quad (\varepsilon > 0).$$

A's unit cost falls *below* B's by 0.1ε, letting A undercut B, capture more market share, raise Q_A further, and lower costs again—**circular causation** that drives the industry toward *complete specialization* in one country.

(d) Trade between identical countries. Even if A and B are identical ex ante, historical accidents (small initial ε) can select a single production location. The specialized country exports to the other at the common world price; thus identical countries *do* trade due to external economies and path dependence (not factor endowments).

(e) Policy and permanent relocation. A temporary production subsidy (or infant-industry protection) in A raises Q_A, lowers C_A, and can tip the industry into A. After scale builds, the subsidy can be removed; the new low-cost location persists (hysteresis) because exiting the cluster would raise costs.